7.3 Nucleosides

A nucleoside is composed of purine or pyrimidine base and a pentose sugar. In the case of purine nucleoside, the sugar is attached to N-9 of the purine ring, whereas in pyrimidine nucleoside, the sugar is attached to the N-1 of the pyrimidine ring. This type of linkage is N-glycosidic linkage(Fig. 7.4).

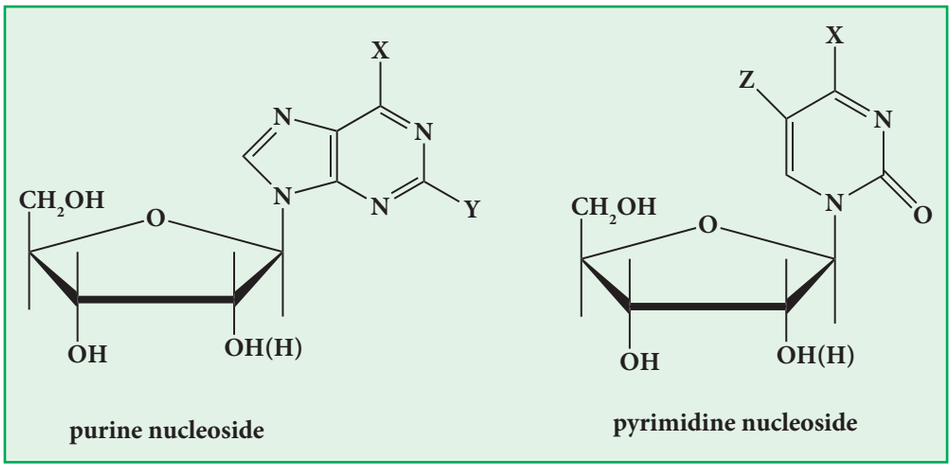


Figure 7.4 Structure of a nucleoside

7.4 Nucleotides

Nucleotides are phosphorylated forms of nucleosides. Phosphorylation (addition of phosphate group) occurs generally in the 5′OH group of the ribose or deoxy ribose sugar. A nucleotide with one phosphate group is called as monophosphate of the respective nucleoside. For example, the monophosphate of adenosine is called as Adenosine Mono Phosphate. If two phosphate groups are attached to the sugar moiety at the 5′OH group, it is called as diphosphate. eg. Cytidine Diphosphate. A nucleotide containing three phosphate groups is called as triphosphate eg. Adenosine Triphosphate (Fig. 7.5). The corresponding nucleosides and nucleotides are listed in Table 1 (Bases, their nucleosides and nucleotides).

Functions

- Nucleotides are the energy currency of the cells (ATP).
- They actively take part in metabolism as hydrogen donors, phosphate group donors and methyl group donors.
- They form the structural components of some co-enzymes (NAD and FAD).

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